

Intense laser solid state physics - unraveling the difference between semiconductors and dielectrics: supplemental material

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DERIVATION OF THE TWO-BAND EQUATIONS

Our analysis is based on a two-band model of a solid using a single valence (v) band and a single conduction (c) band. Here we derive the two-band equations of motion for the probability amplitudes for population in each band.

We begin with the time-dependent Schrödinger equation (TDSE) in the length gauge where the time-dependent Hamiltonian is written as $H(t) = H_0 - \mathbf{x} \cdot \mathbf{F}(t)$, where $H_0 = T + U$ is the unperturbed Hamiltonian with $T = (1/2)\nabla^2$ the kinetic energy and $U(\mathbf{x})$ the periodic potential of the lattice. The field free Hamiltonian H_0 has Bloch eigenstates $\Phi_{m,\mathbf{k}}(\mathbf{x}) = u_{m,\mathbf{k}}(\mathbf{x}) \exp(i\mathbf{k} \cdot \mathbf{x})$ with energies $E_{m,\mathbf{k}} = E_m(\mathbf{k})$ in band $m \in \{v, c\}$ with crystal momentum $\mathbf{k}$; $u_{m,\mathbf{k}}$ is the periodic part of the Bloch function.

In the presence of the laser field the wavefunction becomes time-dependent. In the length gauge it is represented as

$$\Psi(\mathbf{x}, t) = \sum_{m \in \{v, c\}} \int_{\text{BZ}} a_m(\mathbf{k}, t) \Phi_{m,\mathbf{k}}(\mathbf{x}) d^3\mathbf{k} \quad (1)$$

where $a_m(\mathbf{k}, t)$ are the probability amplitudes and integration is over the full Brillouin zone (BZ). To ensure normalization of Ψ , the probability amplitudes are subject to the constraint,

$$\sum_{m \in \{v, c\}} \int_{\text{BZ}} |a_m(\mathbf{k}, t)|^2 d^3\mathbf{k} = 1 \quad (2)$$

for every time t . The derivation that follows is an abbreviated version of the multi-band equations derivation presented in Ref. 1. All equations will be presented in atomic units unless otherwise indicated.

The derivation proceeds by substituting Eq. (1) into the TDSE and integrating out the Bloch eigenstates to yield,

$$\dot{a}_m(\mathbf{k}, t) = [-iE_m(\mathbf{k}) + \mathbf{F}(t) \cdot \nabla_{\mathbf{k}}] a_m(\mathbf{k}, t) + i\mathbf{F}(t) \cdot \mathbf{d}_{mm'}(\mathbf{k}) a_{m'}(\mathbf{k}, t) \quad (3)$$

where $m \neq m'$ and,

$$\mathbf{d}_{mm'}(\mathbf{k}) = i \int d^3\mathbf{x} u_{m,\mathbf{k}}^*(\mathbf{x}) \nabla_{\mathbf{k}} u_{m',\mathbf{k}}(\mathbf{x}) \quad (4)$$

is the transition dipole. To simplify integration of Eqs. (3) we remove the term containing $\nabla_{\mathbf{k}}$ by using the transformation $a_m = b_m \exp(-i \int_{-\infty}^t E_m dt')$ together with $\mathbf{K} = \mathbf{k} - \mathbf{A}(t)$ where $\mathbf{A}(t)$ is the vector potential defined by $\mathbf{F} = -d\mathbf{A}/dt$. As a result we arrive at,

$$\dot{b}_m(\mathbf{K}, t) = i\Omega_{mm'}(\mathbf{K}, t) b_{m'}(\mathbf{K}, t) e^{iS_{mm'}(\mathbf{K}, t)} \quad (5)$$

where $\Omega_{mm'}(\mathbf{K}, t) = \mathbf{F}(t) \cdot \mathbf{d}_{mm'}(\mathbf{K} + \mathbf{A}(t))$ is a K -dependent Rabi frequency and

$$S_{mm'}(\mathbf{K}, t) = \int_{-\infty}^t \varepsilon_{mm'}(\mathbf{K} + \mathbf{A}(t')) dt' \quad (6)$$

is the classical action with $\varepsilon_{mm'} = E_m - E_{m'}$ being the bandgap between bands m and m' . As a result of the transformation to K , the Brillouin (BZ) zone is moving with $\bar{\text{BZ}} = \text{BZ} - \mathbf{A}(t)$.

By defining $\mathbf{d}_{vc}(\mathbf{K}) = \mathbf{d}$, $\Omega(t) = \mathbf{F}(t) \cdot \mathbf{d}$, $\varepsilon(\mathbf{K}) = \mathbf{E}_c(\mathbf{K}) - \mathbf{E}_v(\mathbf{K})$ and $S_{cv}(\mathbf{K}, t) = S(\mathbf{K}, t)$ we can write the two-band equations as,

$$\dot{b}_v(\mathbf{K}, t) = i\Omega(\mathbf{K}, t) b_c(\mathbf{K}, t) e^{-iS(\mathbf{K}, t)} \quad (7a)$$

$$\dot{b}_c(\mathbf{K}, t) = i\Omega^*(\mathbf{K}, t) b_v(\mathbf{K}, t) e^{iS(\mathbf{K}, t)}. \quad (7b)$$

These equations provide the starting point for the derivation of our generalized Keldysh model.

THE GENERALIZED KELDYSH MODEL

In what follows we define $\mathbf{k} = (k, \mathbf{k}_\perp)$ and $\mathbf{K} = (K, \mathbf{k}_\perp)$ with $k = K + A(t)$ and $\mathbf{k}_\perp$ crystal momenta parallel and perpendicular to laser polarization, respectively. We begin by expressing our bandgap as,

$$\begin{aligned} \varepsilon(\mathbf{k}) &= E_g + \varepsilon_\perp(\mathbf{k}) + \varepsilon_\parallel(k) \\ &= \bar{E}_g(\mathbf{k}) + \varepsilon_\parallel(k) \end{aligned} \quad (8)$$

where E_g is the minimum bandgap, $\varepsilon_\perp$ is the general bandgap orthogonal to the laser polarization and $\bar{E}_g = E_g + \varepsilon_\perp$. Along the direction of laser polarization the bandgap is written as,

$$\varepsilon_\parallel(k) = \sum_j^\infty \alpha_j \cos(jka) \quad (9)$$

where a is the lattice spacing which is related to the reciprocal lattice vector $\kappa = 2\pi/a$ along the laser polarization direction. The bandwidth of the material is given by $\Delta = \max[\varepsilon_{\parallel}]$.

Using the trigonometric identity $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$, the bandgap $\varepsilon_{\parallel}[K+A(t)]$ can be written as,

$$\varepsilon_{\parallel}(K+A(t)) = \sum_j^{\infty} \alpha_j \left\{ \cos(jKa) \cos[jaA(t)] - \sin(jKa) \sin[jaA(t)] \right\}. \quad (10)$$

Consider a vector potential of the form $A(t) = -(F_0/\omega_0)f(t/\tau_0)\sin(\omega_0 t)$ where F_0 is the peak electric field strength, ω_0 is the fundamental laser frequency and $f(t)$ is the pulse envelope with width τ_0 . Using the Fourier-Bessel expansion, the trigonometric terms containing the vector potential above become,

$$\cos[jaA(t)] = J_0[j\beta f(t/\tau_0)] + 2 \sum_{n=1}^{\infty} J_{2n}[j\beta f(t/\tau_0)] \cos(2n\omega_0 t) \quad (11a)$$

$$\sin[jaA(t)] = 2 \sum_{n=1}^{\infty} J_{2n-1}[j\beta f(t/\tau_0)] \sin((2n-1)\omega_0 t). \quad (11b)$$

where the J_n 's are the Bessel functions of the first-kind and,

$$\beta = \frac{aF_0}{\omega_0} \quad (12)$$

is the ratio of the Bloch frequency $\omega_b = aF_0$ to the angular frequency of the field. With this we can write the bandgap $\varepsilon = \varepsilon_s + \varepsilon_{ns}$ as the sum of the sinusoidal terms (ε_s) and the non-sinusoidal terms (ε_{ns}); these are expressed as,

$$\varepsilon_{ns}(\mathbf{K}, t) = \bar{E}_g(\mathbf{K}) + \sum_{j=0}^{\infty} \alpha_j \cos(jKa) J_0[j\beta f(t/\tau_0)] \quad (13a)$$

$$\begin{aligned} \varepsilon_s(\mathbf{K}, t) = & 2 \sum_{j=0}^{\infty} \alpha_j \left\{ \cos(jKa) \right. \\ & \times \sum_{n=1}^{\infty} J_{2n}[j\beta f(t/\tau_0)] \cos(2n\omega_0 t) \Big\} \\ & - 2 \sum_{j=0}^{\infty} \alpha_j \left\{ \sin(jKa) \right. \\ & \times \sum_{n=1}^{\infty} J_{2n-1}[j\beta f(t/\tau_0)] \sin((2n-1)\omega_0 t) \Big\}. \end{aligned} \quad (13b)$$

This allows us to write the classical action as $S = S_{ns} + S_s$. With this, Eq. (7b) can be written as,

$$\dot{b}_c(\mathbf{K}, t) = iL(K, t)e^{iS_{ns}(\mathbf{K}, t)} \quad (14)$$

where

$$L(\mathbf{K}, t) = \Omega^*(\mathbf{K}, t)b_v(\mathbf{K}, t)e^{iS_s(\mathbf{K}, t)}. \quad (15)$$

Expressed in terms of its Fourier transform, $L(K, t)$ is given as,

$$L(\mathbf{K}, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{L}(\mathbf{K}, \omega) e^{-i\omega t} d\omega. \quad (16)$$

In order to connect with Keldysh theory, we split $L(K, t)$ into contributions from frequency bands of width σ to get,

$$L(\mathbf{K}, t) = \sum_{n=-\infty}^{\infty} L_n(\mathbf{K}, t) e^{-in\sigma t} \quad (17)$$

where,

$$L_n(\mathbf{K}, t) = \frac{1}{\sqrt{2\pi}} \int_{-\sigma/2}^{\sigma/2} \tilde{L}(\mathbf{K}, n\sigma + \omega) e^{-i\omega t} d\omega. \quad (18)$$

By replacing $\tilde{L}(\mathbf{K}, \omega)$ with its inverse Fourier transform, $L_n(\mathbf{K}, t)$ can further be expressed as,

$$\begin{aligned} L_n(\mathbf{K}, t) &= \frac{1}{2\pi} \int_{-\sigma/2}^{\sigma/2} d\omega \int_{-\infty}^{\infty} dt' L(\mathbf{K}, t') e^{i(n\sigma + \omega)t'} e^{-i\omega t} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} dt' L(\mathbf{K}, t') e^{in\sigma t'} \frac{e^{i\frac{\sigma}{2}(t-t')} - e^{-i\frac{\sigma}{2}(t-t')}}{i(t-t')} \\ &= \frac{\sigma}{2\pi} \int_{-\infty}^{\infty} dt' L(\mathbf{K}, t') e^{in\sigma t'} \text{sinc} \left[\frac{\sigma}{2\pi}(t-t') \right] \end{aligned} \quad (19)$$

where $\text{sinc}(x) = \sin(\pi x)/(\pi x)$. Finally, we define the conduction band population through

$$n_c(t) = \int_{\text{BZ}} d^3K |b_c(\mathbf{K}, t)|^2 \quad (20)$$

and use Eqs. (14), (15) and (17) to write it as,

$$n_c(t) = \int_{\text{BZ}} d^3K \left| \sum_{n=-\infty}^{\infty} \int_{-\infty}^t dt' L_n(\mathbf{K}, t') R_n(\mathbf{K}, t') \right|^2 \quad (21)$$

where R_n is defined as,

$$R_n(\mathbf{K}, t) = e^{iS_{ns}(\mathbf{K}, t) - in\sigma t}. \quad (22)$$

To obtain the generalized Keldysh model we first make the frozen valence band approximation $b_v(\mathbf{K}, t) \approx b_v(\mathbf{K}, t = -\infty) = b_0$. Here it is assumed that the valence

band dynamics have little influence on transition to the conduction band. Next we assume that envelope $f(t_s/\tau_0)$ has negligible change over a cycle and can be taken as constant (quasi-continuous wave approximation); here t_s is a slow varying time. By letting $\sigma = \omega_0$ and considering only the t -dependence of the rapidly varying sinusoidal functions, L_n becomes the Fourier integral,

$$L_n(\mathbf{K}, t_s) = \frac{\omega_0}{2\pi} \int_{-T_0/2}^{T_0/2} dt' L(\mathbf{K}, t_s, t') e^{in\omega_0 t'}. \quad (23)$$

The time integral in Eq. (21) can then be written as,

$$\int_{-\infty}^t L_n(\mathbf{K}, t') R_n(\mathbf{K}, t') dt' = L_n(\mathbf{K}, t_s) \int_{-\infty}^t R_n(\mathbf{K}, t') dt'$$

where the quasi-continuous wave approximation has allowed for L_n to be moved outside the time integral. As $t \rightarrow \infty$, the remaining time integral can be evaluated as,

$$\begin{aligned} & \lim_{t \rightarrow \infty} \left| \int_{-\infty}^t R_n(\mathbf{K}, t') dt' \right|^2 \\ &= \lim_{t \rightarrow \infty} \left| \int_{-\infty}^t e^{i\varepsilon_{ns}(\mathbf{K}, t_s) t' - in\omega_0 t'} dt' \right|^2 \\ &= \lim_{t \rightarrow \infty} \frac{4 \sin^2 \left((\varepsilon_{ns}(\mathbf{K}, t_s) - n\omega_0) t/2 \right)}{(\varepsilon_{ns}(\mathbf{K}, t_s) - n\omega_0)^2} \\ &= 2\pi t \delta(\varepsilon_{ns}(\mathbf{K}, t_s) - n\omega_0). \end{aligned} \quad (24)$$

Equation (21) then becomes,

$$n_c(t) = 2\pi t \int_{\text{BZ}} d^3K \sum_{n=-\infty}^{\infty} |L_n(\mathbf{K}, t_s)|^2 \delta(\varepsilon_{ns}(\mathbf{K}, t_s) - n\omega_0). \quad (25)$$

The only terms that will contribute to ionization are those for which the argument of the δ -function in Eq. (25) is equal to zero. This gives the conditions for the n -photon resonant channels that satisfy energy conservation as,

$$\bar{E}_g(\mathbf{K}) + \sum_{j=0}^{\infty} \alpha^j \cos(jKa) J_0[j\beta f(t_s/\tau_0)] = n\omega_0, \quad (26)$$

where ε_{ns} has been expanded using Eq. (13a). Equation (26) can then be solved for $K = K_n(\beta, \mathbf{k}_\perp, t_s)$. Real values of K_n represent ionization, that is, the positions in K -space where there will be conduction band population at the end of the laser pulse.

With the resonance condition in Eq. (26), integration over K in Eq. (25) can be performed. Using the δ -function condition, the ionization rate is then given as,

$$\begin{aligned} \frac{dn_c}{dt_s} &= 2\pi \int_{\text{BZ}} d^3K \sum_{n=-\infty}^{\infty} |L_n(\mathbf{K}, t_s)|^2 \frac{\delta(K - K_n)}{|\varepsilon'_{ns}(\mathbf{K}_n, t_s)|} \\ &= 2\pi \int_{\text{BZ}_\perp} d\mathbf{k}_\perp \sum_{n \in N_r} \frac{|L_n(\mathbf{K}_n, t_s)|^2}{|\varepsilon'_{ns}(\mathbf{K}_n, t_s)|} \end{aligned} \quad (27)$$

where $\mathbf{K}_n = (K_n, \mathbf{k}_\perp)$, N_r is the set of all n -photon resonant channels, $\varepsilon'_{ns} = d\varepsilon_{ns}/dK$ and $L_n(\mathbf{K}_n, t_s)$ can be written as,

$$\begin{aligned} L_n(\mathbf{K}_n, t_s) &= \frac{\omega_0}{2\pi} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} dt' L(\mathbf{K}_n, t_s, t') e^{i\varepsilon_{ns}(\mathbf{K}_n, t_s) t'} \\ &= \frac{\omega_0 b_0}{2\pi} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} dt' \Omega^*(\mathbf{K}_n, t_s, t') e^{iS(\mathbf{K}_n, t_s, t')}. \end{aligned} \quad (28)$$

where the resonance condition in Eq. (26) and $L(\mathbf{K}, t)$ in Eq. (15) has been used when going from the first line to the second line in Eq. (28). Note that Eq. (27) applies to any bandgap and depends on time only through $f(t_s/\tau_0)$. Further, depletion is not contained in our GK calculations, but can be added by multiplying the right hand side of Eq. (27) with n_c [2]. Comparison of original Keldysh and GK results (not shown here) for the calculations in Figs. 1(a) and (d) of the main manuscript shows good agreement everywhere except in the dielectric where channel closing dominates the ionization dynamics.

Finally, the integral $d\mathbf{k}_\perp$ can be performed approximately by noting that the dependence on $\mathbf{k}_\perp$ is dominantly exponential [3]. We define

$$g_n(\mathbf{k}_\perp, t_s) = \frac{|L_n(\mathbf{K}_n, t_s)|^2}{|\varepsilon'_{ns}(\mathbf{K}_n, t_s)|} \quad (29)$$

and set $g_n = \exp(h_n)$. We then Taylor expand $h_n = \ln(g_n)$ to second order about $\mathbf{k}_\perp = 0$. Evaluation of the resulting Gaussian integral yields

$$\frac{dn_c}{dt_s} = 2\pi \sum_{n \in N_r} g_n^{(0)} \left(\frac{2\pi}{\sqrt{|\hat{M}_n|}} e^{\frac{1}{2} \mathbf{m}_n^T \hat{M}_n \mathbf{m}_n} \right), \quad (30)$$

Here, $g_n^{(0)} = g_n(0, t_s)$ and

$$\mathbf{m}_n^i = \partial_{k_\perp^i} h_n \Big|_{\mathbf{k}_\perp=0} \quad (31a)$$

$$\hat{M}_n^{ij} = -\partial_{k_\perp^i} \partial_{k_\perp^j} h_n \Big|_{\mathbf{k}_\perp=0} \quad (31b)$$

with subscript n referring to $K = K_n$ and $k_\perp^{i,j}$ referring to the individual components of $\mathbf{k}_\perp = (k_\perp^1, k_\perp^2)$. For one dimensional systems the term in parentheses in Eq. (30) is equal to unity. As such, Eq. (30) reduces to $dn_c/dt_s = 2\pi \sum_{n \in N_r} g_n^{(0)}$ for one dimensional systems.

THE EFFECT OF DRIVING FIELD WAVELENGTH ON CHANNEL CLOSING

To demonstrate the effect that a longer driving field wavelength will have on optical field ionization, we consider the generalized Keldysh model at two different values of ω_0 for the dielectric used in the main manuscript.

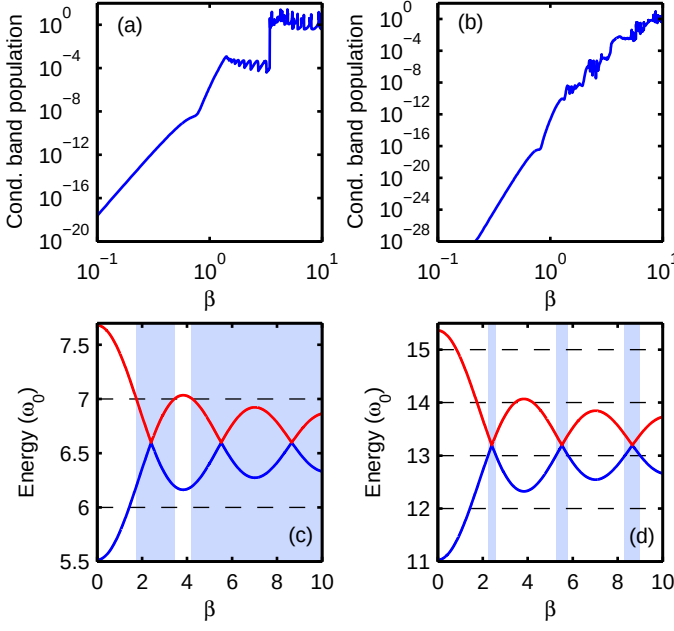


FIG. 1: Top row: Final conduction band population versus β for the dielectric calculated from the generalized Keldysh model for (a) $\omega_0 = 0.06$ a.u. and (b) $\omega_0 = 0.03$ a.u. Bottom row: Demonstration of channel closing for (c) $\omega_0 = 0.06$ a.u. and (d) $\omega_0 = 0.03$ a.u. The solid lines give the maximum (red) and minimum (blue) of the left hand side of Eq. (26). The dashed lines give the allowed n -photon transitions and the shaded regions indicate the range of β for which all channels are closed.

Figures 1(a) and 1(b) show final conduction band population versus β calculated from the generalized Keldysh model for $\omega_0 = 0.06$ a.u. (same as Fig. 1(a) in the main manuscript) and $\omega_0 = 0.03$ a.u., respectively. When comparing the two curves, it is clear that the longer wavelength in Fig. 1(b) has a behavior much similar to that seen in the semiconductor. Figures 1(c) and 1(d) demonstrate the reason behind this behavior. The solid lines indicate the maximum (red) and minimum (blue) of the left hand side of Eq. (26) and the dashed lines give the n -photon resonances. When the dashed line is above the maximum or below the minimum that n -photon resonance channel is closed. As discussed in the manuscript, conditions can arise where all n -photon channels are closed for a range of β . The ranges where all channels are closed are represented by the shaded regions. For $\omega_0 = 0.06$ a.u. there are only two possible channels; for $\omega_0 = 0.03$ a.u. there are twice as many. The increased density of channels within the bandwidth greatly reduces the ranges of β over which all n -photon resonance channels are closed. This then causes the optical field ionization in the dielectric to behave similarly to that in a semiconductor where many channels are supported.

SUMMARY OF VARIABLES

	Variable	Description
Band structure and material parameters	$\mathbf{k}$	Crystal momentum $\mathbf{k} = (k, \mathbf{k}_\perp)$
	k	Crystal momentum $\parallel$ to laser polarization
	$\mathbf{k}_\perp$	Crystal momentum $\perp$ to laser polarization
	$\mathbf{K}$	$\mathbf{K} = (K, \mathbf{k}_\perp)$
	K	$K = k - A(t)$
	$\varepsilon(\mathbf{k})$	Bandgap between valence and conduction bands
	$\varepsilon_\parallel(k)$	Bandgap $\parallel$ to laser polarization
	$\varepsilon_\perp(\mathbf{k})$	Bandgap $\perp$ to laser polarization
	E_g	Minimum bandgap energy
	$\bar{E}_g(\mathbf{k})$	$E_g + \varepsilon_\perp(\mathbf{k})$
Laser pulse parameters	Δ	Bandwidth ($\max[\varepsilon_\parallel]$)
	a	Lattice constant along $\parallel$ direction
	α_j	Fourier coefficients for $\varepsilon_\parallel$
	$\mathbf{F}(t)$	Laser electric field
	$\mathbf{A}(t)$	Vector potential
	F_0	Peak electric field strength
	ω_0	Circular frequency of field
	τ_0	FWHM of laser field
	$f(t)$	Pulse envelope
	t_s	Slowly varying (envelope) time
Ionization models	$b_{v,c}$	Probability amplitudes for valence/conduction bands
	$n_{v,c}$	Population for valence and conduction bands
	$\mathbf{d}$	Dipole
	Ω	$\mathbf{F} \cdot \mathbf{d}$
	S	Classical action
	$S_{s,ns}$	Sinusoidal/non-sinusoidal contributions to $S = S_s + S_{ns}$
	$\varepsilon_{s,ns}$	Sinusoidal/non-sinusoidal contributions to $\varepsilon_\parallel = \varepsilon_s + \varepsilon_{ns}$
	β	$F_0 a / \omega_0$
	$L(\mathbf{K}, t)$	$\Omega^*(\mathbf{K}, t) b_v(\mathbf{K}, t) e^{iS_{ns}(\mathbf{K}, t)}$
	$\tilde{L}(\mathbf{K}, \omega)$	Fourier transform of L

TABLE I: Summary of the terms and variables used in the main manuscript.

The terms and variables used in the main manuscript are summarized in the table above. This is intended to assist the reader by providing a summary of the variables

in one place. However, this is not to be taken as a replacement for the full definitions provided in the main manuscript (or those provided above).

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